Open-Source Edge LLMs for Forecasting Stock Returns via News Headline Sentiment Analysis

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I certify that this dissertation, and the research to which it refers, are the result of my own work.

Abstract

This research investigates the capability of open-source Large Language Models (LLMs) to forecast stock returns using news headline sentiment analysis, with a specific focus on edge models deployable on consumer-grade hardware. The Llama3-8B family of models was evaluated, including quantized versions and a fine-tuned FinGPT variant, against benchmark models (FinBERT, GPT-3.5, GPT-4o mini, and DeepSeek V3) as well as existing LLM solutions from current research (FinLlama, FinLLaMa, StockGPT and ChatGPT-3.5). Using a Kaggle dataset of US financial headlines from October 2018 to December 2019, we showed edge LLMs able to outperform some of the larger, generalized LLMs in classification and portfolio performance. Overall accuracy remains moderate, reflecting the complexity of market prediction and challenges of obtaining high quality news datasets. The largest quantized model, Llama3-8B (INT8), yielded impressive portfolio returns (42.4% annualized, 1.55 Sharpe ratio), performing competitively with existing FinLLMs and significantly outperforming the S&P 500 benchmark. Just as impressive, it delivered this performance while running within 10GB of RAM on standard consumer hardware (M2 Macbook Pro with 16GB RAM). These findings highlight the potential of small, resource-efficient open-source LLMs to power profitable trading strategies, democratizing access to advanced AI techniques. At the same time, we demonstrate that research in this area is in need of further work, as the current landscape suffers from lack of consistent study design, transparency of metrics and peer review of results.

Keywords: Large language models, edge models, open source, finance, news sentiment analysis, algorithmic trading

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Chapter 1

Introduction

1.1 Background

For more than a decade, the financial industry has seen increasing adoption of machine learning and deep learning techniques to optimize trading strategies. According to a 2017 assessment by Morgan Stanley, algorithmic trading strategies managed approximately $1.5 trillion across hedge funds, mutual funds, and smart beta ETFs (Jansen 2018). Reporting by The Economist indicates a few investment firms, including Two Sigma and Renaissance Technologies, have eliminated human traders entirely, relying exclusively on machine learning for trading decisions ('The stock market is now run by computers, algorithms and passive managers' 2019).

However, these fully automated approaches remain uncommon. Research from Ferreira et al. (2021) shows that among companies utilizing algorithmic trading, approximately 90% continue to employ predefined, hard-coded procedures rather than true artificial intelligence. This suggests that AI applications in financial trading are still in early stages and could expand significantly if profitable and economically viable approaches are developed.

The introduction of OpenAI's ChatGPT two years ago, and the proliferation of other large language models (LLMs) since, has dramatically transformed the landscape, bringing transformer networks and LLMs to prominence. This development has catalyzed renewed enthusiasm for artificial intelligence's potential across both academic and commercial sectors.

LLMs represent a significant advancement in natural language processing (NLP), creating valuable opportunities for financial institutions. As highlighted in the review by Ashtiani and Raahemi (2023), news content has emerged as a valuable information source for stock market prediction. Multiple studies have demonstrated the efficacy of these textual sources in forecasting equity prices. This approach aligns with fundamental analysis principles, which maintain that stock price movements reflect changes in company financial health, operational performance, and future prospects. News articles provide critical insights for evaluating a company's fundamental outlook.

Lopez-Lira and Tang (2023) demonstrated impressive trading performance from supplying ChatGPT-3.5 with news headlines and asking it to output sentiment labels. Without providing it any additional domain knowledge in finance, the model yielded highly profitable portfolio returns over a large variety of stocks.

Recently, domain-specific models have begun to emerge with expertise and contextual understanding of finance, often referred to as "FinLLMs." Bloomberg developed its proprietary BloombergGPT for internal use in forecasting, news analysis, and various financial tasks. This specialized 50 bil-

lion parameter LLM was trained on Bloomberg's exclusive internal datasets, requiring 53 days of training at an approximate cost of $3 million (Wu et al. 2023).

In response to BloombergGPT, various open-source FinLLMs have been developed in the last two years, such as FinGPT (H. Yang et al. 2023), FinLlama (Konstantinidis et al. 2024) and the similarly named FinLLaMa (Xie et al. 2024). These are each based on Meta's open-source Llama model, which has itself rapidly progressed in performance and capabilities. That progress can be seen in the Llama3 series. Through versions 3.1, 3.2 and 3.3, progressively smaller versions of the model have been released, down to a mere 1 billion parameters. These edge models are beginning to demonstrate capabilities previously possible only with much larger variants.

1.2 Motivation

The progress of open-source LLMs, both general purpose models and specialized FinLLMs, represents a significant opportunity to democratize advanced trading strategies previously accessible only to institutional investors with access to large amounts of private financial data and substantial computational resources. However, critical questions remain about the performance and practical applicability of these models in real-world trading environments. In particular, the question has not yet been explored for edge models running on widely accessible, consumer-grade hardware.

1.3 Aims

This research is motivated by three primary aims.

First, the author seeks to conduct a comparative evaluation of open-source LLMs and FinLLMs against general-purpose base models to quantify their relative performance in stock price prediction tasks. This comparison will seek to demonstrate whether open-source models can match the performance of proprietary models, and if domain-specialized models can best general-purpose alternatives, when applied to financial news sentiment analysis and trading strategy.

Second, the author aims to investigate the viability of models small enough to run on consumer hardware, made possible by model evolution and architecture advances such as quantization and LoRA fine-tuning. This addresses the practical economic question of whether retail investors and smaller financial firms can leverage these advanced AI tools to develop profitable trading strategies without requiring enterprise-level computational infrastructure.

Third, differing metrics, stock markets, stock types and trading simulation strategies have been employed in existing benchmarks of LLM trading performance, sometimes making direct comparison difficult. This research will publish a fuller comparison of LLM metrics than has previously been done, lending to greater transparency and confidence in this research area.

Chapter 2

Literature Review

2.1 Forecasting markets

Competing theories exist over whether markets are random or predictable. The efficient market hypothesis (EMH) holds that any new information about a stock is immediately reflected by the markets in its price, and so it is functionally impossible to have an "edge" over the market (Malkiel and Fama 1970). More recently, the adaptive market hypothesis (AMH) has gained prominence. It states that as markets are driven by people, they are subject to some of the same irrationality as people are, presenting opportunities for savvy investors to profit when the market has mispriced an asset (Lo 2005).

Forecasting financial markets can lead to significant profitability but remains a complex challenge. The two dominant approaches to stock market modeling are fundamental analysis and technical analysis. Fundamental analysis evaluates a stock's value by primarily examining a company's strengths and weaknesses. Its balance sheet, debt obligations, and so on are examined, as well as larger economic factors such as the health of its sector and the larger macroeconomic environment (Beyaz et al. 2018). In technical analysis, only a stock's chart and price history are taken into account. Various indicators are calculated based on price history, such as moving averages, MACD, RSI and others, and a stock's prospects are determined by how its price compares to these indicators (Anbalagan and Maheswari 2015).

In fundamental analysis, news headlines are a primary tool as they not only indicate potential future developments for a company, but have been shown to hold a positive correlation with subsequent stock performance (J. Zhang et al. 2018). Deveikyte et al. (2022) demonstrated that analyzing headlines alone revealed a statistically significant connection to next-day returns in the FTSE 100. Curiously, attempts to analyze the main text of news articles did not show similar correlation with subsequent market movements, nor did sentiment analysis of Twitter content.

2.2 Evolving NLP techniques

Natural language processing (NLP) and text mining have become essential techniques for extracting information from unstructured textual data. These include social media posts, news articles, and financial reports, which contain rich but often difficult-to-analyze information. Earlier statistical approaches used text representation methods such as term frequency (TF), which primarily analyzes the statistical characteristics of text (Hagenau et al. 2013). While these methods provided

a foundation for text analysis, they have significant limitations, particularly the inability to capture the contextual meaning and semantic relationships between words.

To address these shortcomings, the field has evolved toward more sophisticated learning-based models. These include Word2vec, GloVe, and fastText, which represented a significant advancement in capturing word semantics by embedding words in a continuous vector space (Goldberg and Levy 2014). This evolution continued with the development of more advanced language models, notably BERT (Bidirectional Encoder Representations from Transformers). BERT represented a revolutionary approach as it considers both backward and forward word directions simultaneously, allowing it to develop a richer understanding of context (Devlin et al. 2018). This bidirectional approach enabled BERT to grasp nuanced meanings and relationships between words that earlier models could not.

Building on BERT, Araci (2019) released FinBERT, addressing the unique domain knowledge present in financial documents. Its contribution to the field was substantial, outperforming existing language models like ELMo and ULMFit when these were pre-trained for financial tasks, with an accuracy of 0.86 and F1 Score of 0.84 when evaluated on the Financial PhraseBank and TRC2-financial datasets, the latter being a database of financial news articles.

We now see momentum shifting toward transformer-based models, primarily due to OpenAI's release of the GPT models and ChatGPT interface in recent years. The transformer architecture first emerged in the groundbreaking publication "Attention is All You Need" (Vaswani et al. 2017) which introduced an innovative attention mechanism. This advancement not only enabled effective pattern tracking by the network regardless of when patterns appeared in the sequence, but significantly, allowed computation to occur in parallel, enabling dramatically larger models to be trained with equivalent hardware.

LLMs have shown highly sophisticated sentiment analysis capabilities. However, the field of stock market prediction is just beginning to take advantage of them (Ashtiani and Raahemi 2023). Various papers published within the past two years have attempted to fill this gap.

Shobayo et al. (2024) compared modern NLP models (FinBERT and GPT-4) against traditional machine learning (logistic regression) for the task of predicting stock market movements from news headlines. They found logistic regression outperformed both FinBERT and GPT-4 with an impressive 82% accuracy score, while GPT-4 yielded only 54%, leading to the conclusion that while LLMs excel in the task of understanding sentiment in complex or nuanced text, fine-tuned or domain specific models can still outperform them, even those using substantially less computational resources.

Y. Shen and P. K. Zhang (2024) benchmarked GPT-4o against FinBERT for analyzing financial sentiment of news articles, financial reports and company announcements. By fine-tuning FinBERT and providing few-shot prompting to GPT-4o, they were able to achieve accuracy scores of 88% and 86%, respectively.

Lopez-Lira and Tang (2023) demonstrated success providing headlines and a related company to ChatGPT, prompting the model to forecast how the headline would impact the next trading day's price action for the given stock. The model was asked to classify the headline as positive, negative or neutral for the company in the short term. Their results indicated that a GPT-3.5 strategy purchasing stocks with a positive score and short-selling stocks with a negative score would have yielded a cumulative return between October 2021 and December 2022 exceeding 550%, with a Sharpe ratio of 3.09. Interestingly, the more advanced GPT-4 model yielded smaller returns at 350% but a higher Sharpe ratio of 3.8. This excludes transaction costs, as is standard practice in most

related research. This outcome is notable not only in the high profitability, but also considering they used "zero-shot" prompting; no examples were provided in the prompt.

2.3 Open-source FinLLMs

BloombergGPT was the first FinLLM released, trained specifically with financial data for financial applications (Wu et al. 2023). Being an internal Bloomberg initiative, however, its broader impact has been limited. Open-source alternatives are desirable for many reasons, which the AI4Finance group articulated effectively:

- Advancing equal opportunities via democratizing FinLLMs
- Cultivating transparency and trust
- Accelerating research and innovation
- Enhancing education
- Promoting community development and collaborative engagement

In the last couple years, various FinLLMs based on Meta's open-source LLaMa models have been developed. The AI4Finance group released the first paper demonstrating the potential of open-source financial LLMs. They developed FinGPT on top of the Llama2-7B foundation model and enhanced it through Low-Rank Adaptation (LoRA), which optimizes trainable parameters while minimizing computational demands.

In the last year, two more implementations for financial applications have emerged bearing the same name. (Konstantinidis et al. 2024) released FinLlama, adapting the Llama2-7B foundation model for finance, implementing parameter-efficient fine-tuning through LoRA. When evaluated in portfolio management scenarios, this model demonstrated impressive performance, outperforming the established FinBERT baseline by 44.7% in cumulative returns while achieving higher Sharpe ratios and lower annualized volatility. The authors demonstrated that fine-tuning a large language model with even a small amount of task-specific data can yield strong results.

Later that same year, Xie et al. (2024) released their own FinLLaMa model, this one based on the updated Llama3-8B foundation model. Pre-trained on a 52 billion token financial corpus including text, tables, and time-series data, it outperforms baseline models across financial tasks. Their instruction-tuned variant, FinLLaMA-instruct, demonstrated superior performance in sentiment analysis, named entity recognition, and number understanding, outperforming GPT-4 in these areas.

StockGPT takes a different approach, functioning as an autoregressive numeric model rather than a traditional language model. Trained on a dataset of 70 million daily U.S. stock returns spanning nearly a century, StockGPT starts with custom decoder-only transformer architecture and treats return series as token sequences, leveraging attention mechanisms to identify predictive patterns. With an equal-weighted portfolio strategy, it achieved very impressive 119% annualized return on data from 2001 - 2023 and a Sharpe ratio of 6.3. (Mai 2024)

2.4 Resource-efficient strategies

Three key approaches have emerged in recent years which enable performant yet small LLM architectures: quantization, knowledge distillation, and LoRA.

Quantization reduces model precision by converting weights from higher-precision formats (32-bit floating point or FP32) to lower-precision formats (8-bit integers or less). This signific-

antly decreases memory requirements while maintaining reasonable performance. As Frantar et al. (2023) demonstrated, optimized quantization techniques like GPTQ can compress models by approximately 4x compared to FP16 with minimal quality degradation.

Model distillation is a technique to enhance small model performance by transferring knowledge from larger "teacher" models to smaller "student" models. Hinton et al. (2015) introduced this approach, showing that students can learn to mimic the output distributions of teachers rather than just their final predictions. More recently, Sanh et al. (2019) demonstrated with DistilBERT that students can retain 97% of performance while using only 40% of the parameters of their teachers.

LoRA (Low Rank Adaptation) offer an efficient approach to fine-tuning by freezing pre-trained model weights and injecting trainable rank decomposition matrices into specific layers. Hu et al. (2022) found this method enables fine-tuning with just 1% of the parameters needed for full fine-tuning. This dramatically reduces memory requirements while preserving adaptability to new tasks.

Chapter 3

Methodology

As noted above, current research benchmarking FinLLM performance suffers due to differing approaches to trading simulation, as well as different stock markets and even stock type being tested. This could erode confidence in the capabilities of open-source LLMs. To address this issue, this research largely follows the methodology implemented by Lopez-Lira and Tang (2023). Their research provided the greatest detail on the trading simulation used and is therefore highly replicable. By applying their approach in this research, the reader can trust in the comparisons presented later in this paper. In addition, we publish additional metrics like accuracy and F1 score for all models tested in our portfolio simulation, bringing greater transparency to how these models function.

3.1 Data

While web scraping live headlines is an ideal approach to data acquisition, as it ensures headlines were not part of the model's training data, this introduces implementation complexities and ethical considerations regarding publication terms of service. This research made use of the publicly available Kaggle dataset "Daily Financial News for 6000+ Stocks" (Aenlle 2018). Not only did this provide sufficient headlines across a variety of news sources and company stocks, but it also included stock ticker labels indicating the company referenced by the headline. This labeling was essential, since the task given the model was not overall sentiment analysis of the headline, but sentiment analysis for a given company.

Lopez-Lira and Tang (2023)'s results were based on a selection of headlines from October 2019 to December 2020. As the Kaggle dataset did not extend to December 2020, an alternate 18-month period of October 2018 to December 2019 was used in its place. By drawing from the same start month and end month, any seasonality differences were controlled for. The author also considers it preferable to avoid the COVID time period. The extreme economic conditions exhibited during that time raises the question of how generalizable the results are.

The use of headlines from October 2018 to December 2019 presents a data contamination concern. Most modern LLMs, including open-source models, were likely trained on datasets that include financial news from this period. This could artificially inflate performance metrics, as the models might be recalling specific headline-market correlations rather than demonstrating genuine predictive capabilities.

This concern is largely mitigated by the fact that that news publication dates will not be provided to the model, making it extremely unlikely the model would remember any correlation between

a headline to its related stock performance. Only one day of data will be provided to the model at a time before comparison to the following day's stock price. In addition, each prompt will be a completely separate API call, preventing any context from being developed.

Further, this research primarily focuses on small models, which have much smaller memorization capabilities (e.g. 8 billion parameters for Llama3-8B compared to an estimated 1.7 trillion parameters for GPT-4). Any possible correlation of stock movements and news headlines in the model training data would advantage the larger models over the smaller ones. Finally, time and resource constraints did not allow for novel data collection through webscraping or professional news aggregation services. This meant that use of any publicly available dataset would also come with this consideration, as none of the suitable news headline datasets were recent enough to be outside the training periods for the Llama3 or the other LLMs.

Unlike some of the results published in this domain, where performance was demonstrated on only a handful of well-known stocks, this dataset contained 699 stocks in the sample period, representing a diversity of company sizes and sectors. The data was enriched with additional stock information so it could be filtered to those companies listed on the American Stock Exchange (AMEX), New York Stock Exchange (NYSE), or the National Association of Securities Dealers Automated Quotations (NASDAQ). Additional filtering was done to only include share codes 10 or 11, which represent ordinary common stocks, following the practice of similar studies.

Whereas Lopez-Lira and Tang (2023) were able to filter articles based on a relevance score provided by RavenPack, indicating the relevance of a given news item to a company, this was not possible with the Kaggle dataset. Indeed, the dataset included thousands of headlines which contained no information whatsoever about a given company, such as "46 Stocks Moving In Friday's Mid-Day Session", "9 Stocks To Watch For February 18, 2020", and "Earnings Scheduled For May 21, 2020". These headlines only added noise to the dataset, providing little to no information on which to assess sentiment.

To denoise these types of headlines, this research adopted a simple strategy. A headline not containing either a company's stock ticker or the company's common name would be filtered out. Stock tickers containing only 1 letter were removed from the dataset to accommodate this filtering. This ensured not only that headlines like the above were eliminated, but also that the remaining headlines would be highly relevant to a company.

We further simplified the dataset by removing headlines that were highly similar. The Restricted Damerau-Levenshtein distance metric was used, a string metric that quantifies the dissimilarity between two sequences by calculating the minimum number of operations required to transform one string into another. Unlike the standard Levenshtein distance, which only considers the first three operations, the Damerau-Levenshtein variant also accounts for transpositions of adjacent characters, making it effective for identifying headlines that contain the same information but with slight syntactic variations. Headlines for the same company on the same day with a similarity score exceeding 0.6 were removed from the analysis. This ensured results were not biased by over-representation of essentially identical news items.

After data preparation, 59,257 headlines remained in the dataset. Testing showed that using the slowest API of the group, DeepSeek V3, this would take roughly 138 hours to process. Sampling was therefore employed to make inference time manageable. Stratified sampling was employed, which organized the dataset into month-year strata, maintaining proportional representation of stocks across the entire time period. See Table 3.1.

Table 3.1: News Headline Dataset

	Raw	Filtered & Denoised	Stratified Sample
Time Period	2/2009-6/2020	10/2018-12/2019	10/2018-12/2019
Headlines	1,399,180	59,257	2,995
Trading days	3,957	324	312
Stocks	6,192	2,090	699
Inference time (DeepSeek V3)	3,269 hours	138 hours	7 hours

3.2 Model Selection

Llama3-8B will serve as the base model for testing. LLaMa ("Large Language Model Meta AI") remains the leading open-source LLM project and has been employed in the majority of FinLLM studies over the past few years. It will be tested in its full precision form as well as at INT8, INT6 and INT4 quantization levels. For fine-tuned models, FinGPT will be tested, using its QLoRA adapter on top of Llama3-8B. FinGPT was selected because it was the only available FinLLM which has been updated for Llama3. The others were either not available at the time of this study or were based on the older Llama2 architecture.

Newer Llama3.1 models were available at the time of this study. However, the FinGPT QLoRA adapter was designed to run on top of Llama3-8B, and for this reason Llama3-8B was treated as the base model across all testing. This approach ensures when comparing results between the base model and the fine-tuned variant, changes in performance will only be due to the fine-tuning layer. For further benchmarking, DeepSeek V3, GPT 3.5 and 4o-mini and FinBERT will also be tested.

Fine-tuning via LoRA adapter will be the primary strategy for providing domain specialization to the model. Retrieval Augmented Generation (RAG) is a common solution for this as well, as discussed above. However, due to the need to batch process thousands of headlines in this research, in contrast to live processing of new headlines from web scraping, RAG would introduce unacceptable latency.

Inference for full resolution, full-size models was done via calls to provider APIs. Due to the large amounts of memory required by these models, API calls provide the most cost-efficient approach. In contrast, quantized Llama3-8B inference can be run on consumer-level hardware. A Macbook Pro laptop with M2 Pro processor and 16GB RAM will be utilized for local inference. llama.cpp will be employed to serve the models, as it provides the most optimized utilization of Apple's Metal graphics APIs. For example, Llama3-8B in the quantized INT8 variant, the largest model which could run locally, consumes approximately 10GB of memory with llama.cpp.

In classical machine learning, model parameters are tested and refined on a subset of test data from the overall dataset, to determine the best performing model settings for a given problem. As LLMs are so much larger than traditional ML algorithms, often several orders of magnitude lager, the model training step is often replaced by prompt engineering, setting a small set of hyperparameters, and/or techniques such as LoRA fine-tuning. For the purposes of this study, the temperature hyperparameter was set to 0 to ensure the highest possible consistency of results between runs of a given model. Details on the prompt testing and design process follow.

3.3 Prompt design

Lopez-Lira and Tang (2023)'s prompt instructed the model to output "YES" for positive news for a company, "NO" for negative news for a company, and "UNKNOWN" if the sentiment could not be determined. These outputs were then converted to 1, -1 and 0, respectively, and then averaged for a given stock across all headlines for the day to determine a final trade score. If the average trade score was positive, a long trade was indicated. If the average score was negative, a short trade was indicated. And if the average score was 0, then no trade was indicated.

This approach was improved upon in this study. As LLMs possess nuanced sentiment analysis capabilities, a "YES" response for all positive sentiments is unnecessarily simplistic. The model can gauge not just if a headline is positive for a company, but the degree of how positive it is. In this research, the model was prompted to output scores from -1 to 1 in increments of 0.25. This improved precision becomes particularly important when a company has multiple headlines in a given day, where multiple scores which could be positive or negative are averaged together. The overall positivity or negativity of the day's headlines is much better captured in this approach.

The rest of the prompt design was improved upon as well, either by removing unnecessary instructions or by providing a clearer framework to the model. For example, "Forget all previous instructions" was unnecessary, as each headline was fed in a new API call to the model, eliminating any concerns about context being retained between headline predictions. Each headline was a new session, which guaranteed the prediction would not be affected by previous or subsequent responses, regardless of whether a given headline was the first, the hundredth or the thousandth provided to the model.

Some of the most cutting edge LLMs today - OpenAI's o1 and o3-mini and DeepSeek R1 - are examples of reasoning models, which demonstrate a breakthrough in performance by employing Chain-of-Thought (CoT) as part of every request. These models "think" aloud, or output reasoning as the response is being formulated. A study by Kojima et al. (2022) demonstrated that even simple prompts like "Let's think step by step" could trigger CoT reasoning, improving performance by as much as 4x times on arithmetic and commonsense reasoning benchmarks.

The prompt used in this research directed the model to provide a one-sentence explanation for its response after providing its score, following Lopez-Lira and Tang (2023)'s approach. It was confined to a single sentence and a maximum token limit was set, for the sake of efficiency. Not only has this been found to enhance the accuracy of model outputs, but adds a valuable explainability dimension to each of the model's predictions. One of the shortcomings of many deep models is the "black box" nature of them - the lack of explainability for how a prediction was generated, which is particularly problematic in organizations when decisions are informed by such models. By adding this crucial level of explainability, this design becomes appropriate in and adoptable by organizations which have strict model risk management policies.

Experimentation was also done with one-shot and few-shot prompting, which provides sample inputs and outputs in the prompt to demonstrate appropriate output. This was particularly important while testing the FinLLaMa-instruct model, as it failed to follow the below guidelines on response output. Even when provided few-shot prompting, however, it continued to fail to follow the required format. Therefore, FinLLaMa was not included in this research, nor was few-shot prompting found to improve output of the other models.

The above prompt experimentation was carried out on a random sample of 957 headlines from 2015 to 2017, outside the testing period of 2018-2019. This approach ensures the final predictions

in this research will be out-of-sample and will demonstrate how well the prompt generalizes to other headline data. Llama3-8B at full resolution, ChatGPT-3.5 and DeepSeek V3 were used as test models.

The following prompt was found to not only produce the most consistent output across models, but also to provide the best accuracy among the prompts tested.

> You are analyzing this headline about {company} as an experienced stock analyst.
> Your goal is to give recommendations that lead to a profitable trade.
> Score the headline's market impact, considering:
> - Most news has minimal real impact (use 0)
> - Consider market expectations vs reality
> - A negative sentiment should yield a negative score
> Use these scores only: -1, -0.75, -0.5, -0.25, 0, 0.25, 0.5, 0.75, 1
> Headline: {headline}
> Respond only in this format:
> SCORE: [number]
> REASON: [one sentence explanation]

The first line directs the model to act as an experienced stock analyst. When prompted to adopt the role of domain experts such as mathematicians or scientists, LLMs exhibit improved reasoning capabilities, producing more methodical and accurate solutions for complex problems. This technique, commonly referred to as "role prompting" or "persona-based prompting," which can be seen in the first part of the prompt, has been demonstrated to activate relevant knowledge encoded within the model's parameters, suggesting that role prompting may help focus the model's attention on pertinent information stored in its weights.

3.4 Trading simulation

As noted above, a wide variety of companies were sampled for the trading simulation. 699 stocks across industries, company sizes and exchanges (NYSE, AMEX or NASDAQ) will be tested. This will avoid the bias that may have been present in other studies which only benchmarked against well-known, large US technology stocks.

Various approaches can be taken in selecting a trading simulation. This is an important dimension of the study because it can have significant impacts on the results and the conclusions drawn in relation to existing research. The trading simulation largely follows Lopez-Lira and Tang (2023)'s excellent methodology. This was a strong approach because it enters and exits a trade as quickly as possible after an article is released, minimizing the possibility for other factors to impact stock movement as can happen when stocks are held multiple days.

Given that back-testing is employed on historical stock data, useful pricing data is limited to market open and close prices. An article published before 9:30AM (or after 4:30PM the prior day) will lead to an open trade at the next market open and will be closed at the next market close. Articles released during the trading day lead to a trade opened at market close and then closed at the next market open.

This represents a small improvement over Lopez-Lira and Tang (2023)'s approach, which treated 6:00AM as the cutoff time, without justification as to why. In other words, an article released at 8:00AM would not be traded at market open, but at market close because it missed the 6:00am

cutoff, Then it would be closed at 9:30AM the following day, more than 24 hours after the article was released. This is not a sound approach when the impact of that news would have almost certainly been reflected in the 9:30AM open the day it was released.

Three portfolio trading strategies will be simulated. The long strategy purchases stocks with a positive trade score. The short strategy shorts stocks with a negative trade score. The long-short strategy will perform both. If a company has multiple headlines for the same trading period, these will be averaged together before a trade decision is made for the day. Finally, the trading portfolio will be equal-weighted, dividing up funds equally among all trades for a day.

3.5 Evaluation metrics

Accuracy and F1 score will be used for the trade score generated on each headline. Despite being a numeric value, these primarily indicate either a positive, negative or neutral trade direction. Thus, this was essentially a classification problem. In calculating accuracy and F1 score, an average trade score for a company for the day was treated as correct if it matched the sign of the subsequent stock move - a positive trade score and a positive stock return or a negative trade score and negative stock return.

FinBERT was also tested as an additional reference point. It assesses overall sentiment of a phrase and outputs one of three sentiment values: 0 for negative, 1 for neutral or 2 for positive. These were converted to trade scores of 1, 0 and -1, respectively.

Cumulative return, annualized return, annualized volatility and Sharpe ratio will be used to assess the trading simulation itself, after a company's headlines are averaged together. These four metrics are widely used in this research area and will enable effective comparison with existing research. They are calculated as follows:

$$\text{Cumulative Return} = \prod_{i=1}^{n}(1 + r_i) - 1 \tag{3.1}$$

$$\text{Annualized Return } R_p = \left(\prod_{i=1}^{n}(1 + r_i)\right)^{\frac{252}{n}} - 1 \tag{3.2}$$

$$\text{Annualized Volatility } \sigma_p = \sqrt{\frac{\sum_{i=1}^{n}(r_i - \bar{r})^2}{n-1}} \times \sqrt{252} \tag{3.3}$$

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p} \tag{3.4}$$

where n is the number of trading days in the period, r_i is the daily return i (expressed as a decimal), $\bar{r}$ is the mean daily return over the period, R_f is the risk-free rate (e.g., treasury bill yield), considered to be 0 in line with other research, as rates were quite low during this period, and 252 is the standard number of trading days in a year.

Chapter 4

Results

4.1 Classification Results

In assessing the results, we will first evaluate sentiment labeling (e.g. the trade scores), before moving on to the portfolio results.

Accuracy and F1 scores are shown in Table 4.1. These metrics are not well reported in current FinLLM research for trading scenarios, and so a broad comparison with existing work is not possible. Y. Shen and P. K. Zhang (2024) demonstrated impressive accuracy and F1 scores for FinBERT and GPT-4o, albeit on a different dataset. They evaluated the Financial Phrasebank which is comprised of contents of news articles. A sentence within an article is likely more simple to classify than a headline. Headlines often convey multiple meanings in a short statement whereas sentences typically have only one primary meaning. For that reason their work was not considered comparable.

All of our results are provided here to bring greater transparency to research in this domain as well as provide more context for the trading performance results.

Table 4.1: Model Classification Performance

Model	Accuracy	Precision	Recall	F1 Score	Total Samples
FinBERT	0.475	0.488	0.423	0.453	1637
GPT-3.5	0.518	0.519	0.622	0.566	2930
GPT-4o-mini	0.521	0.520	**0.673**	0.587	2949
DeepSeek-v3	0.521	0.518	0.622	0.565	2560
Llama3-8B Full	0.527	0.528	0.638	0.578	2798
Llama3-8B_Q8	**0.529**	**0.528**	0.666	**0.589**	2442
Llama3-8B_Q8 QLoRA	0.526	0.527	0.648	0.581	2618
Llama3-8B_Q6	0.525	0.526	0.614	0.567	2781
Llama3-8B_Q4	0.523	0.523	0.663	0.585	2854

Generally, scores are noticeably lower than what is commonly reported in financial sentiment analysis research. This is an expected result, given that there is not a 100% correlation between news headlines and a company's subsequent stock movement. Indeed, Shobayo et al. (2024) reported GPT-4 having an accuracy of only 54% in a similar task, in line with our results.

FinBERT can be seen underperforming the LLMs. This again demonstrates the enhanced ability of LLMs for the complex sentiment analysis task of classifying headlines. Without having a way to instruct FinBERT to predict sentiment for the given company, it could only provide a general

sentiment output for each headline.

Several interesting patterns can be seen among the LLMs. First, all performed within the 51-52% accuracy range and 56-59% for F1 scores. The INT8 quantized version of Llama3-8B narrowly achieved the best overall performance with an accuracy of 0.529 and F1 score of 0.589, outperforming much larger commercial models from the GPT family and DeepSeek V3. This is an unexpected finding, given the much smaller size (8 billion parameters) of Llama3-8B compared to the others. GPT-3.5 is believed to be 175 billion parameters while DeepSeek V3 is the largest of the group at 671 billion parameters. Even highly compressed Llama3 versions delivered competitive results.

Recall scores show greater variation (0.614-0.673), indicating differences in LLMs' abilities to identify relevant positive trading signals. GPT-4o-mini achieved the highest recall (0.673), though its overall accuracy and F1 score were lower than Llama3-8B_Q8.

Another unexpected finding is that the Llama3-8B_Q8 model with FinGPT QLoRA adapter failed to outperform the base Llama3 model. The difference is not very significant, however, as they both performed within a percentage point on both accuracy and F1 score.

A few explanations could account for the tight range of results and the strong performance of small quantized models compared to the full resolution variants. First, all LLMs demonstrate sufficient capacity at sentiment analysis to perform similarly to each other, regardless of model complexity. Second, despite exhaustive effort to denoise the headline dataset, too much noise may have remained for the models to perform much better than 50%. For example, the Ravenpack subscription utilized by Lopez-Lira and Tang (2023) enabled filtering for only major news sources. This was not possible in the Kaggle dataset used in this study. Finally, proponents of the "random walk" theory of markets might argue this is evidence that markets are more random than predictable.

The best performing model, Llama3-8B, was notably more cautious in its predictions, assigning a score of 0 to 18% of the headlines, much more than any of the other models. This can be seen in its lower count of total samples (any sample assigned a score of 0 was removed from the sample count). It showed much more attention to the statement in the prompt "Most news has minimal real impact (use 0)." The lower-performing GPT family of models rarely assigned a score of 0, seeming to ignore this part of the prompt. Figure 4.1 and Figure 4.2 more fully illustrate the different distribution of trade scores assigned by these models.

This is suggestive of the reason why the small Llama models performed so well and why the FinGPT fine-tuned model did not outperform the general models. Different models may require different prompts to perform best. Despite testing our prompt on a selection of models, the final prompt clearly performed best with the Llama3 family. For the other models to perform their best, including the fine-tuned model, they may need their own optimized prompts.

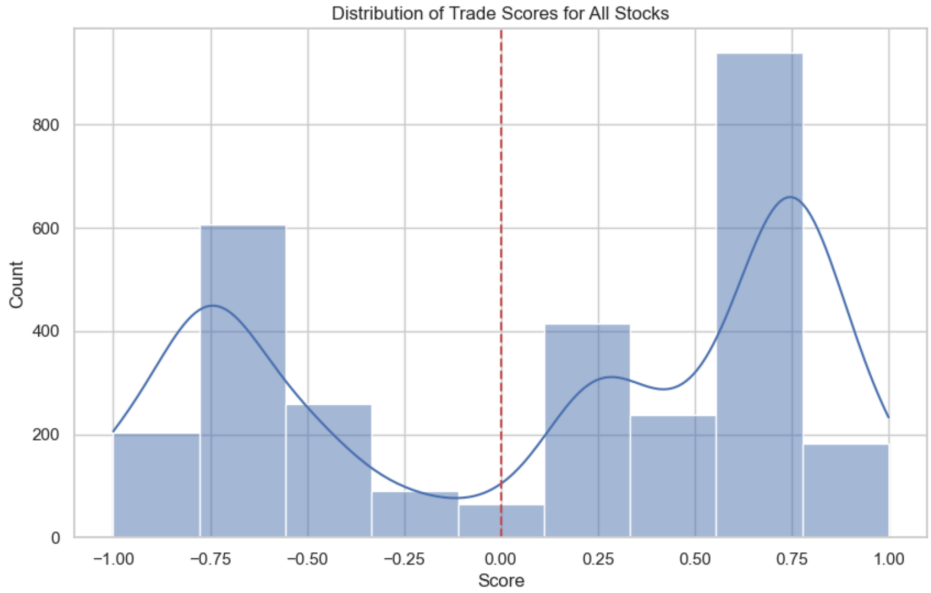

Figure 4.1: GPT-3.5 Score Distribution

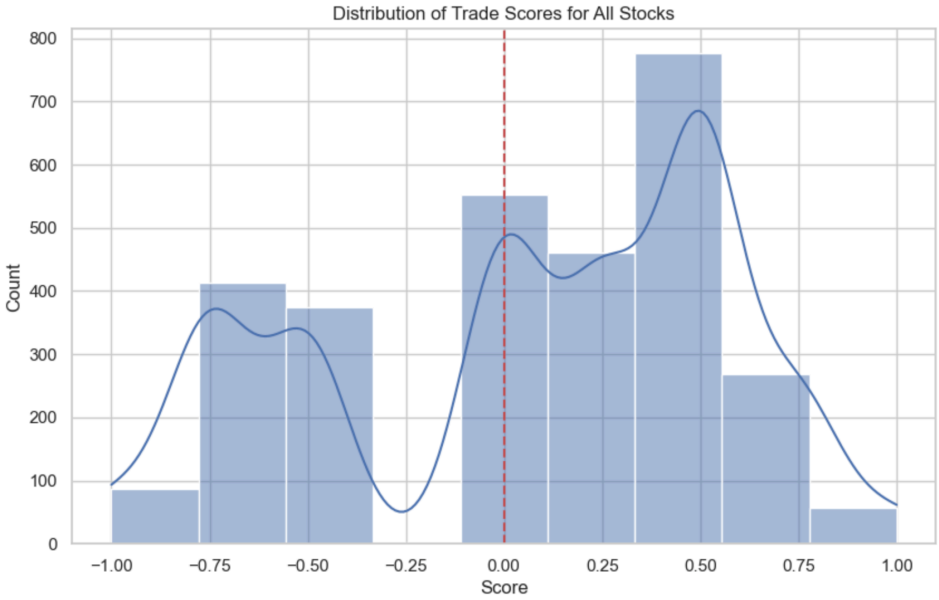

Figure 4.2: Llama3-8B_Q8 Score Distribution

A natural assumption might be the more positive or negative sentiment of a headline, the stronger its predictive power of a profitable long or short trade. Konstantinidis et al. (2024) took this assumption into their simulated trading strategy, only trading companies with the strongest positive or negative headline sentiment scores for the day. This approach was also evaluated in our preliminary testing, but was not found to improve trading performance.

Figure 4.3 and Figure 4.4 suggest why this was the case. They illustrate accuracy and F1 score broken down for each trade score (with scores of 0 omitted from the calculation). Scores are evenly distributed for the most part across sentiment levels. On the positive scale, it can be noted that almost all models showed slightly higher accuracy and F1 scores at the 0.75 trade score. Scores on

the negative scale show a bimodal distribution. Out-sized performance was demonstrated at the -0.25 and again at the extreme of -1.00.

Still, a strategy based on extreme scores would have underperformed for positive news, as the 1.0 trade score showed markedly worse performance across models for both accuracy and F1 score. This could suggest uneven patterns in how LLMs assign numeric sentiment scores to positive items as opposed to negative ones, the need to further tune prompting, or uneven market reactions to strongly positive news as opposed to strongly negative news.

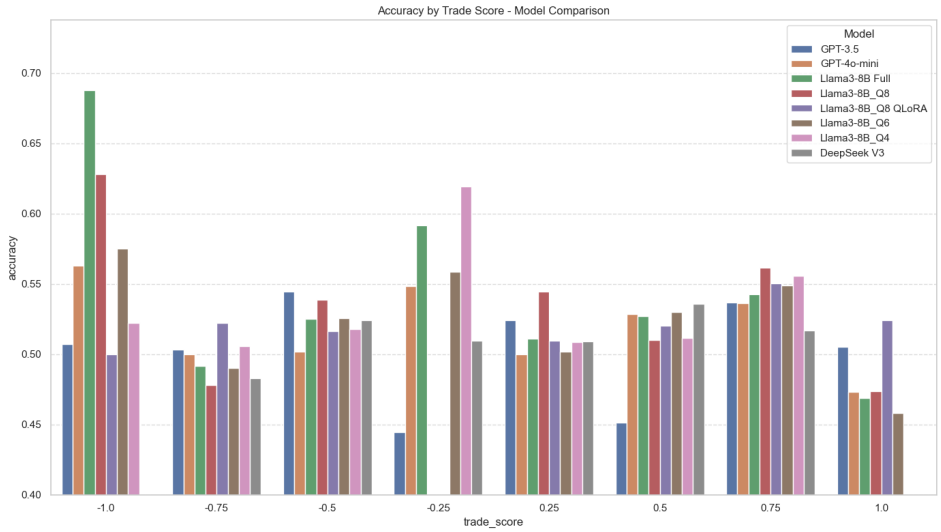

Figure 4.3: Accuracy by Trade Score

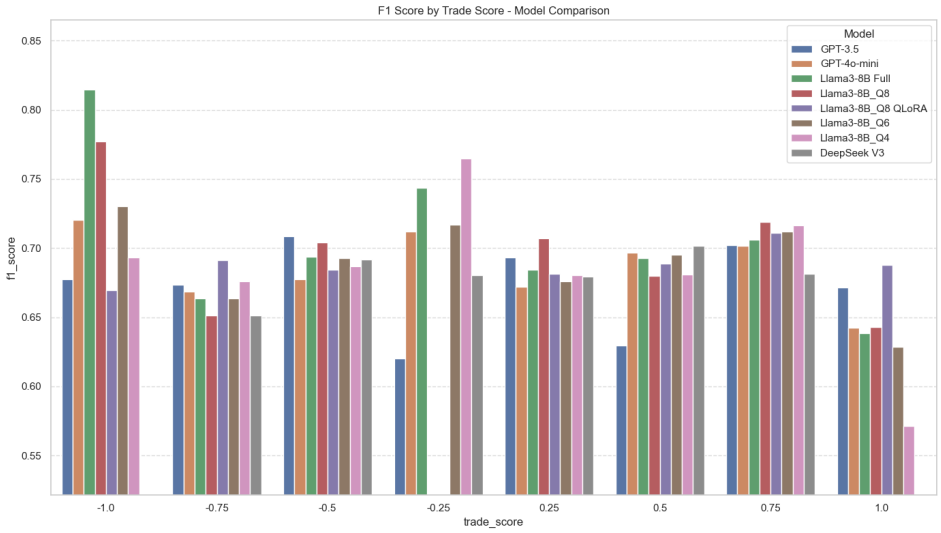

Figure 4.4: F1 by Trade Score

4.2 Portfolio Returns

Three trading strategies were simulated: the long strategy which buys stocks with a positive average trade score for the day, the short strategy which shorts stocks with a negative average trade

score for the day, and the long-short strategy which performs both of the above. The long-short strategy is shown below as it performed best in testing, and it was also used by Lopez-Lira and Tang (2023) and Konstantinidis et al. (2024).

Figure 4.5 shows a boxplot of daily stock returns in the sampled dataset. This shows the data is quite balanced with a median centered around 0, though there are outliers in both directions. This wide range of returns illustrates the challenge of a portfolio simulation - in this case, an incorrect prediction could lead to a 76% loss on a given trade! The 76% movement occurred in PG&E stock (PCG), an example of a major energy company on the NYSE. Outliers such as this were not removed because they represent real-world scenarios on major stocks. Removing them would provide an unrealistic "handicap" to the models.

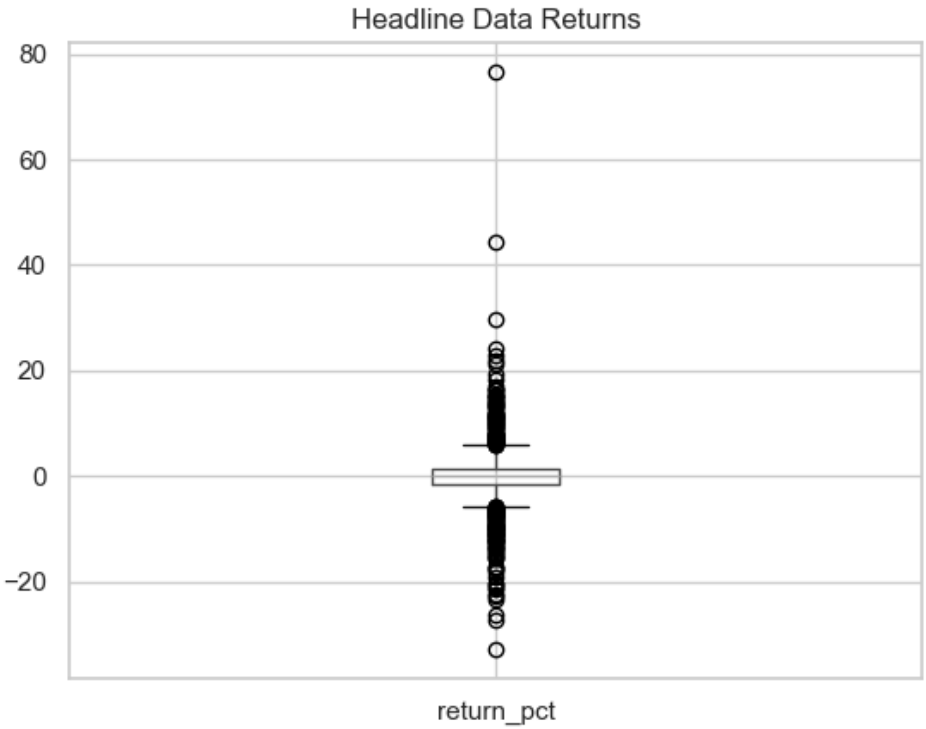

Figure 4.5: Boxplot of Stock Returns in Dataset

Trading results have been split into Table 4.2 and Table 4.3 to highlight the performance of the smaller, quantized local models vs larger comparison models run on servers. Each table includes the S&P500 as a further benchmark.

Model	Cum. Return (%) ↑	Ann. Return (%) ↑	Ann. Volatility (%) ↓	Sharpe Ratio ↑
Llama3-8B Full	**56.68**	**49.65**	28.73	**1.73**
FinBERT	38.65	33.00	**23.95**	1.38
DeepSeek V3	25.13	21.18	28.29	0.75
GPT-3.5	19.73	15.73	30.70	0.51
GPT-4o mini	7.74	5.08	29.98	0.17
S&P 500	19.53	15.35	15.20	0.85

Table 4.2: Benchmark Models Trading Performance

The highest performing model of both groups was the full-precision Llama3-8B model. It

Model	Cum. Return (%) ↑	Ann. Return (%) ↑	Ann. Volatility (%) ↓	Sharpe Ratio ↑
Llama3-8B_Q8	**48.69**	**42.40**	**27.44**	**1.55**
Llama3-8B QLoRA	34.57	29.29	28.93	1.01
Llama3-8B_Q6	32.61	26.65	28.54	0.93
Llama3-8B_Q4	20.48	16.05	30.08	0.53
S&P 500	19.53	15.35	15.20	0.85

Table 4.3: Quantized LLMs Trading Performance

achieved a cumulative return of 56.68%, annualized return of 49.65%, and Sharpe ratio of 1.73. It beat the larger models like DeepSeek and the GPT family, and nearly tripled the cumulative return of the S&P500 over the same period. This is despite not scoring highest for accuracy or F1 scores, which demonstrates that for trading purposes, trading metrics remain the best indicators for modeling real-world results.

All Llama3-8B variants except INT4 outperformed GPT-3.5, GPT-4o mini, DeepSeek V3, and the S&P 500 index in terms of total and annualized returns. Particularly striking is the performance gap between Llama3-8B Full and GPT-4o mini, with the former achieving seven times the total return of the latter (56.68% vs. 7.74%). This is another indication that model size was not the most important factor in accurately and profitably analyzing financial news sentiment. Indeed, this is a promising result and indication of the evolving power of smaller LLMs.

As compression through quantization was introduced (from INT8 to INT4), a consistent decline in returns is observed. The INT8 model still performed impressively, preserving most of the performance (48.69% vs 56.68%), while achieving the lowest volatility (27.44%) across all models. This performance is even more impressive considering it ran inference on only 10GB of memory on a consumer-grade laptop.

Another surprise was that the simpler FinBERT model beat out the FinGPT fine-tuned model, even delivering the lowest annualized volatility of any model tested. Figure 4.6 and Figure 4.7 demonstrate the portfolio performance of each. These show the models taking opposite paths to their returns. While FinGPT made costly mistakes in Q4 2018 and Q1 2019 before recovering, FinBERT achieved most of its returns during that period before declining, demonstrating their prediction tendencies were far more varied than the scores alone suggest. See Appendix A for performance of all tested models.

While the S&P 500 index shows the lowest annualized volatility (15.20%) among all benchmarks, all Llama3-8B variants except INT4 exceeded its Sharpe ratio (0.85). This illustrates that the risk-adjusted returns of our models far surpassed passive index investing during the evaluation period.

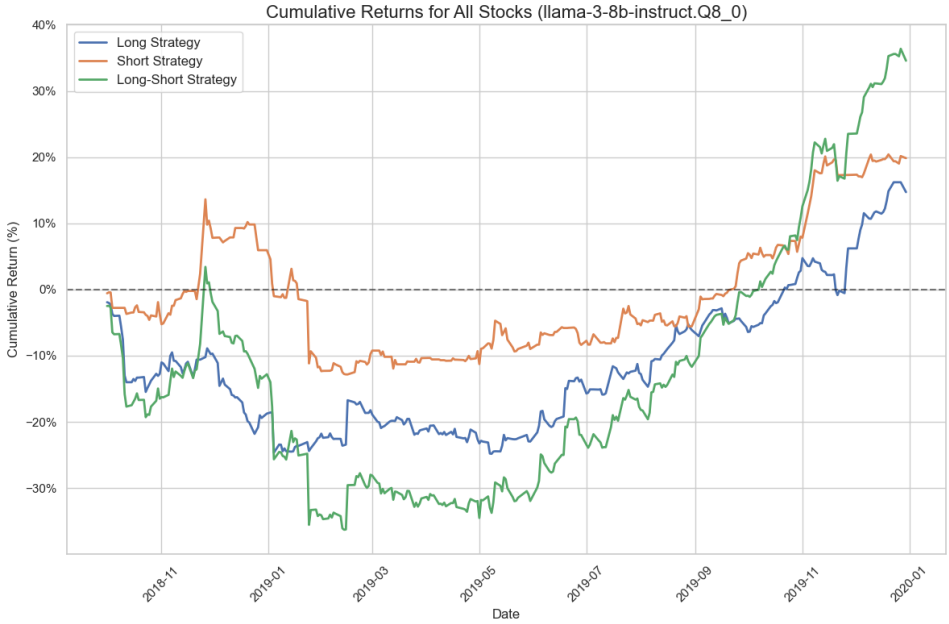

Figure 4.6: FinGPT Portfolio Returns

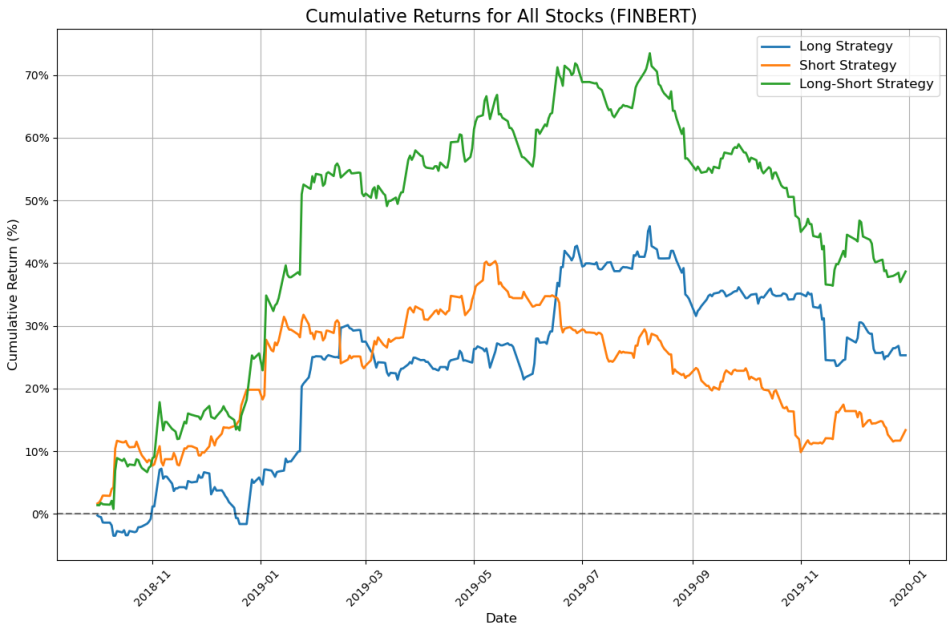

Figure 4.7: FinBERT Portfolio Returns

Table 4.4 shows our work in context of existing research in this emerging area. Konstantinidis et al. (2024)'s FinLlama and Xie et al. (2024)'s FinLLaMa are included along with Lopez-Lira and Tang (2023)'s model. As noted earlier, the former reported a highest performing strategy which returned over 550% in just 15 months, a number almost difficult to believe. They noted this performance relied on a self-financing strategy, however, which no other studies in this area have employed. The norm is an equal-weighted strategy, which our research also followed. Lopez-Lira and Tang (2023) remarked that under an equal-weighted strategy, their cumulative return went from 550%+ to an unreported negative amount, along with a negative Sharpe ratio. To facilitate comparison with

other models which used equal-weighted portfolios, we have used their equal-weighted results (denoted LLT GPT-3.5 EW).

It should be noted others' models were tested on a variety of time periods, making direct comparison difficult. However, other models included here included 2020 data in their testing period. This is during the beginning of COVID, widely considered to be an extreme economic outlier event. For this reason, we suggest our data may generalize better than testing which included the COVID period.

Table 4.4: Model Performance Comparison

Model	**Ann. Return (%) ↑**	**Ann. Volatility (%) ↓**	**Sharpe Ratio ↑**
FinLlama	45.0	**18.6**	**2.4**
FinLLaMa	17.6	47.7	1.4
StockGPT	119	N/A	6.5
LLT GPT-3.5 EW	<0	N/A	<0
Llama3-8B Full	**49.7**	28.7	1.7
Llama3-8B_Q8	42.4	27.4	1.6
FinBERT	33.0	24.0	1.4
Llama3-8B QLoRA	29.3	28.9	1.0
SP500	15.4	15.2	0.9

The table shows promising results from our small Llama3-8B models against existing approaches in financial prediction. Our full Llama3-8B model demonstrates the highest annualized return at 49.65%, outperforming any known models in this field when compared on an equal-weighted portfolio. This impressive performance comes with moderately higher volatility (28.73%) compared to the best performer in the category, FinLlama at 18.6%.

Overall, our quantized Llama3-8B models demonstrate substantial alpha generation compared to the market benchmark while maintaining reasonable risk profiles, with the INT8 model achieving nearly 3x the return of the S&P500 benchmark with less than 2x the volatility. These results show that our open-source edge models can offer significant performance in financial forecasting applications.

Chapter 5

Conclusions

This study sought to evaluate the viability of open-source, edge LLMs in forecasting stock returns through news headline sentiment analysis, while bringing greater transparency to this field by publishing detailed metrics across models. By conducting a comparative analysis of models ranging from current general-purpose LLMs to specialized financial variants, the research provides critical insights into the evolving landscape of AI-driven financial tools.

5.1 Key Findings

The results challenge the assumption that model size is a strong determinant of performance in financial sentiment analysis. The 8 billion parameter Llama3-8B model, both in full resolution and quantized forms, outperformed larger commercial models like GPT-3.5, GPT-4o-mini and DeepSeek V3. Notably, the quantized INT8 variant retained 86% of the full model's performance while operating on consumer-grade hardware, demonstrating that resource efficiency need not compromise performance. This underscores the potential of smaller, optimized models to democratize access to sophisticated trading tools for retail investors and smaller firms.

Domain-specific models like FinBERT and FinGPT underperformed general-purpose LLMs in trading simulations. This suggests that the nuanced reasoning capabilities of modern LLMs may compensate for a lack of financial context when effective prompting strategies.

5.2 Contributions

This research area remains a bit like the "Wild West," notably lacking in peer-reviewed studies or consistency of portfolio simulation. A core contribution of this research and comparative analysis lies in its transparency and reproducibility. By reporting both classification metrics (accuracy, precision, recall, and F1 score) alongside standard portfolio performance metrics (cumulative return, annualized return, volatility, and sharpe ratio), the work addresses a critical gap, offering a fuller picture than often presented. This allows for a more nuanced understanding of model behavior to be developed. Breaking down performance by trade score also provided insights, challenging the assumption that the highest or lowest sentiment scores would yield higher predictive power.

5.3 Limitations and Challenges

The study's reliance on the Kaggle "Daily Financial News of 6000+ Stocks" dataset introduced inherent noise. Despite denoising efforts, the dataset's limitations likely suppressed model performance. The absence of RavenPack-style relevance scoring made it impossible to isolate high-impact news, a factor linked to stronger predictive signals in prior research. Furthermore, potential data contamination could impact metrics, though this risk was mitigated by focusing on edge models with limited memorization capacity.

5.4 Further work

Three avenues for future work emerge from these findings. First, enhancing dataset quality is critical. Incorporating metadata, such as news source size or credibility, could improve signal-to-noise ratios. Additionally, adaptive weighting strategies should be tested. For example, headlines likely have different impacts on a stock depending on the sector, significance of the news source and other factors. Weighting strategies could help improve the performance of models by adjusting for these factors, rather than naively averaging all news headline sentiments together. Finally, enriching the dataset with technical indicators or macroeconomic data could create a more holistic and effective forecasting framework.

Second, optimizing prompts for individual models may unlock further gains. For instance, GPT-family models frequently ignored instructions to assign neutral (0) scores, exhibiting a riskier behavior which cost them in portfolio performance, whereas Llama3-8B adhered more closely to prompt guidelines.

Finally, extending the analysis across different markets, incorporating transaction costs, and eventually moving towards simulated real-time paper trading would provide more robust representations of real-world results.

5.5 Summary

This research demonstrates the potential of modern, open-source, edge LLMs as viable tools for financial sentiment analysis and stock forecasting. By achieving competitive performance with minimal computational resources, models like Llama3-8B challenge the dominance of proprietary systems and pave the way for inclusive innovation. However, the field must address methodological fragmentation and dataset limitations to fully realize this promise. As LLMs continue to evolve, their integration into portfolio workflows, guided by transparency, rigor, and ethical stewardship, could redefine the future of algorithmic trading.

5.6 Disclaimer

This research is presented for academic purposes only. It should not be taken as financial, legal, or investment advice.

5.7 Ethical considerations

Whenever performing news sentiment analysis, the question of copyright and terms of use is raised. Since the research data comes from a publicly available Kaggle dataset, not from web scraping, this question is simplified. The author of the dataset released it under the Creative Commons Zero (CC0) Public Domain license, allowing it to be copied, modified, and distributed without author permission. It is sincerely hoped that the author himself gained permission in compiling the dataset, though this could be not confirmed.

Further ethical considerations lie in the potential impact of this research. If successful, it could encourage broader adoption of AI models for trading purposes. While democratizing access to sophisticated trading methods is a positive outcome, its widespread use could have unpredictable effects on markets. As with all emerging AI technology, supervision and guardrails should be put in place, particularly if AI agents are allowed to perform actions on their own.

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Appendix A

Portfolio Returns by Model

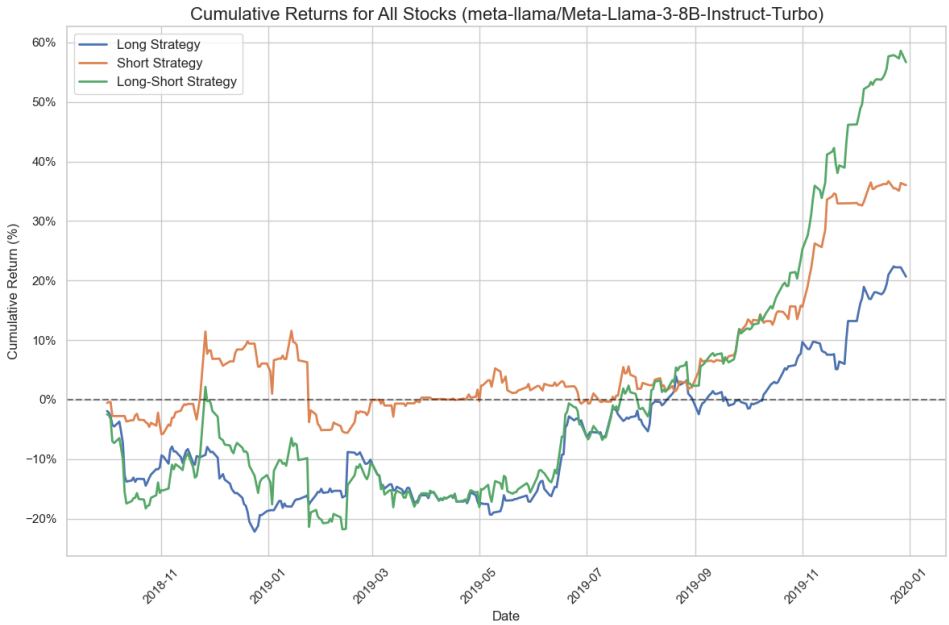

Figure A.1: Llama3-8B Portfolio Returns

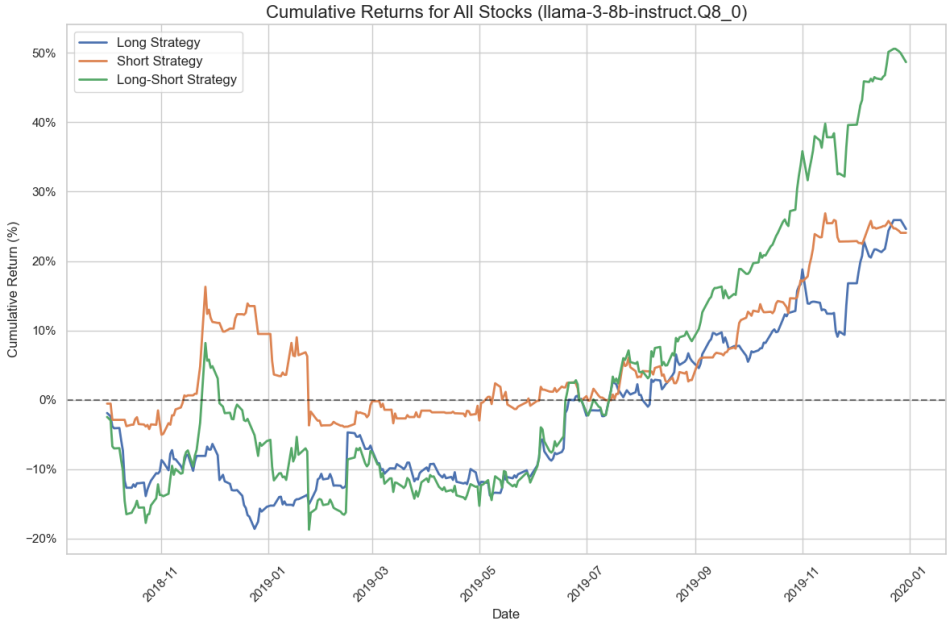

Figure A.2: Llama3-8B (INT8) Portfolio Returns

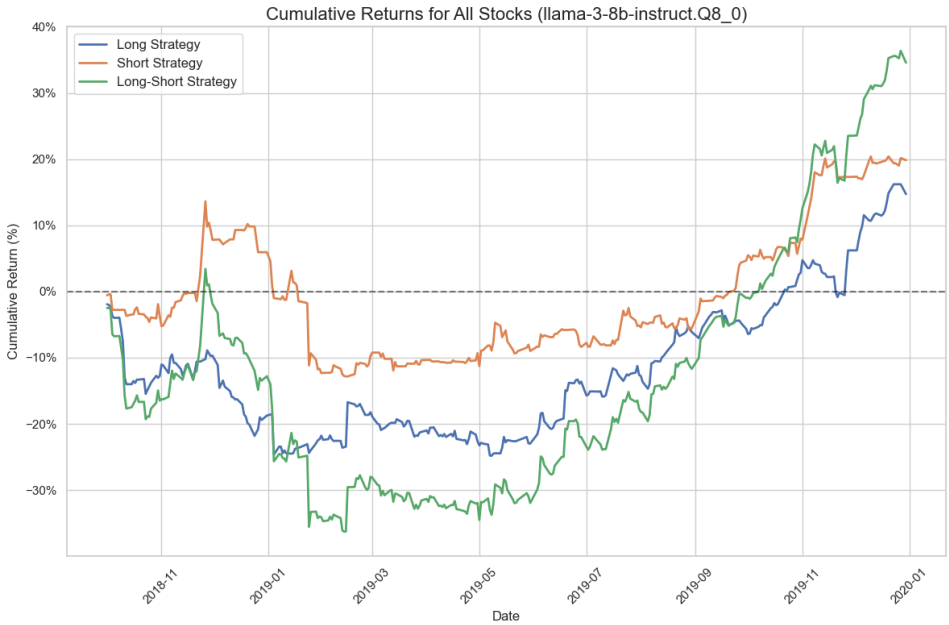

Figure A.3: Llama3-8B QLoRA Portfolio Returns

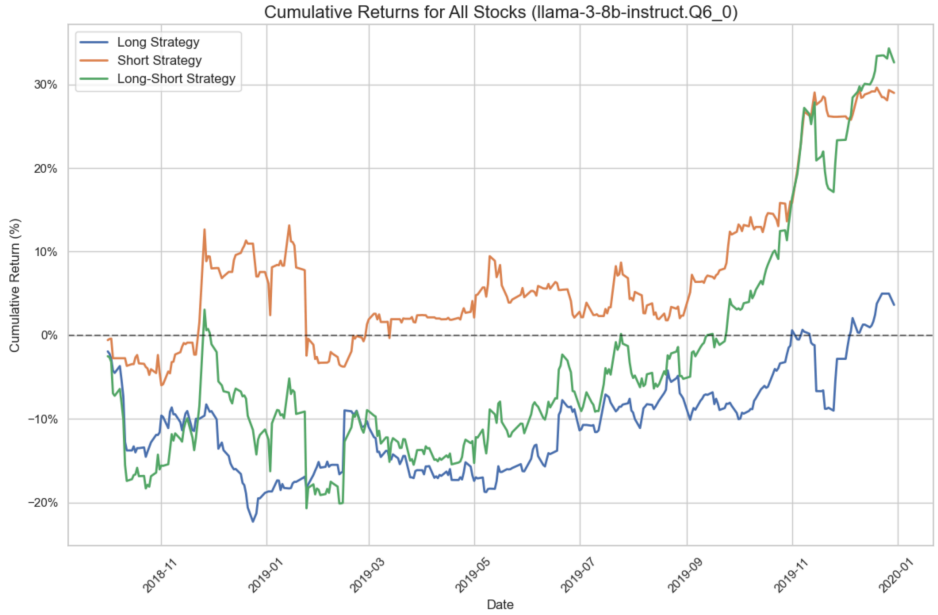

Figure A.4: Llama3-8B (INT6) Portfolio Returns

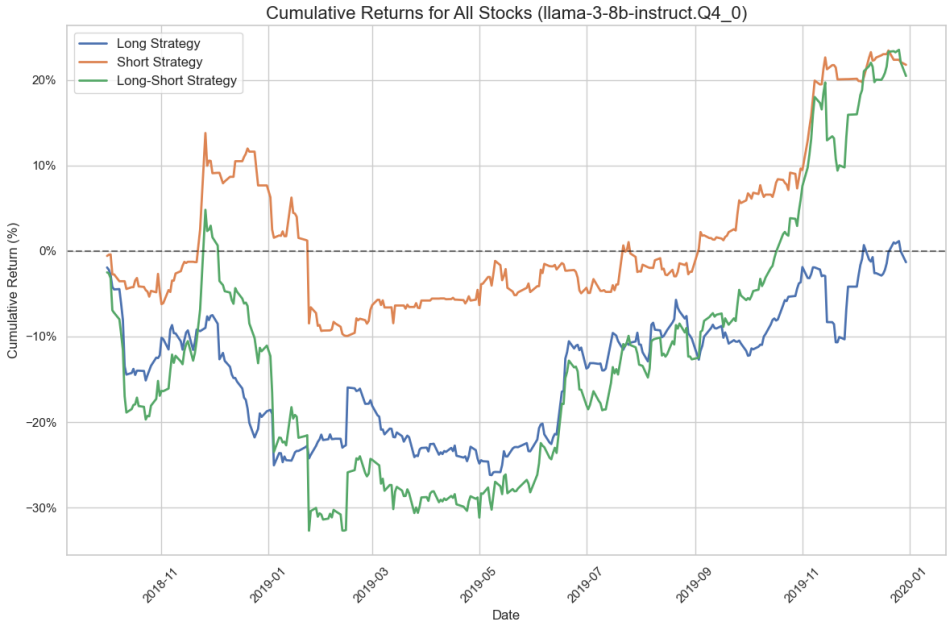

Figure A.5: Llama3-8B (INT4) Portfolio Returns

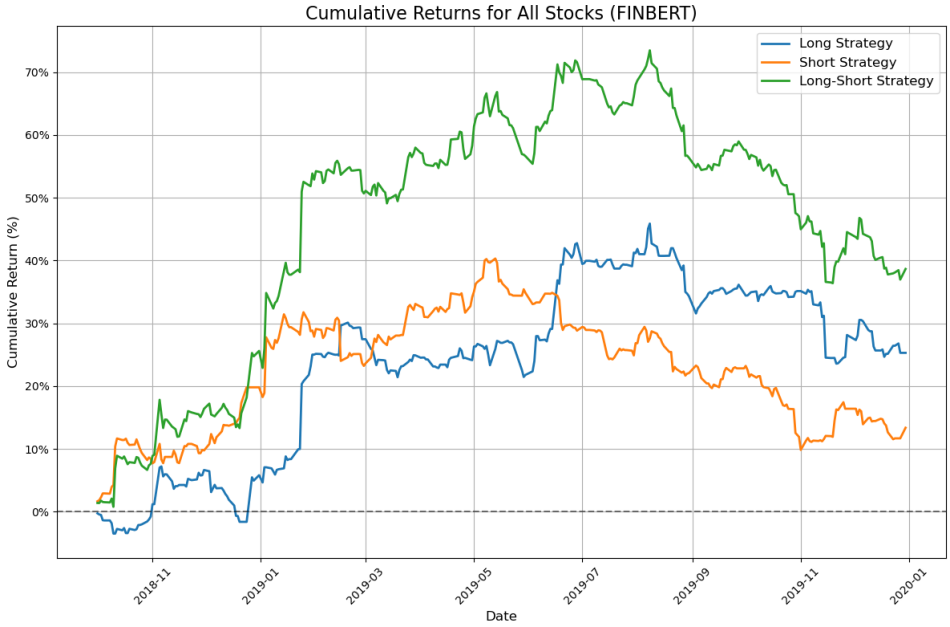

Figure A.6: FINBERT Portfolio Returns

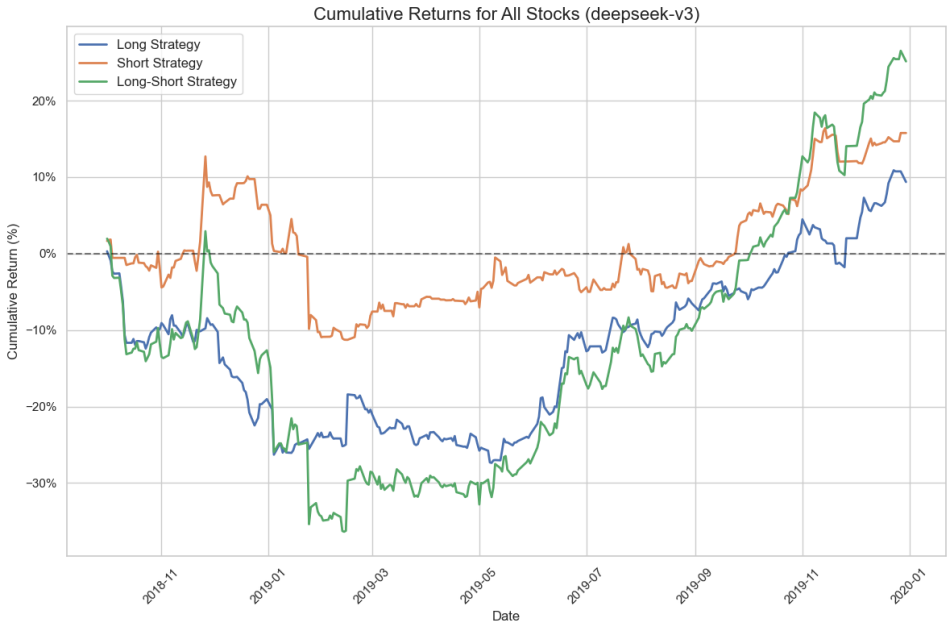

Figure A.7: DeepSeek V3 Portfolio Returns

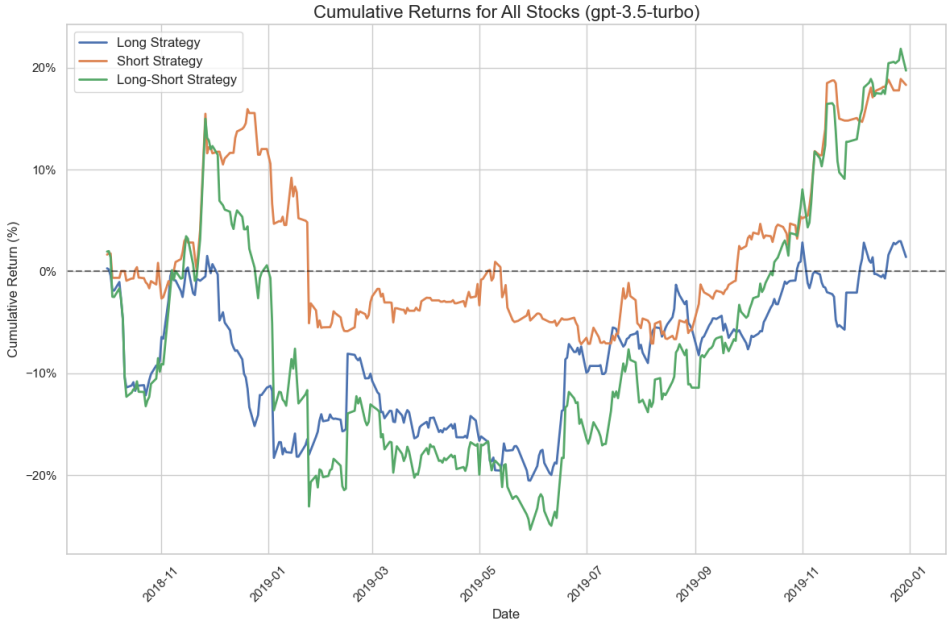

Figure A.8: GPT-3.5 Portfolio Returns

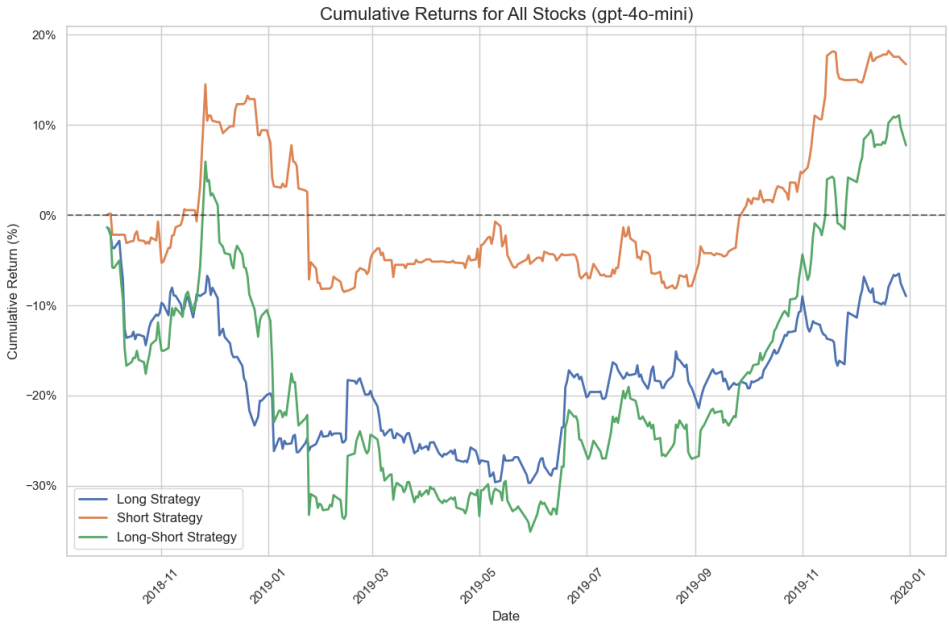

Figure A.9: GPT-4o mini Portfolio Returns

Appendix B

Codebase

The codebase created for this research is available at the below link. Please view the included README for more information and how to run it.

GitHub repository: github.com/colinarndt/dsm500.